

SQL | POWER BI | RISK ANALYTICS

Fraud Detection Dashboard

A comprehensive fraud analytics solution built with SQL & Power BI to detect, analyse, and prioritise high-risk financial transactions.

- by Llinvile de Jongh

18bn

Total Fraud Amount

15.9K

Fraud Cases

0.06%

Recovery Rate

1K

High-Risk Accounts

13M

Total Transactions

PROJECT OVERVIEW

Objective

Design an end-to-end fraud detection pipeline using SQL for data engineering and Power BI for interactive analytics, enabling investigators to prioritise high-risk accounts and identify fraudulent transaction patterns across a 13-million-record dataset.



Data Engineering (SQL)

Cleaned and modelled a 6.3M-row transactions dataset. Built risk-scoring logic, flagged balance mismatches, and ranked accounts by fraud exposure.



Visual Analytics (Power BI)

Designed 4 interactive report pages covering KPIs, fraud type breakdown, timing patterns, balance integrity, and account risk rankings.



Risk Intelligence

Identified CASH_OUT & TRANSFER as sole fraud vectors, surfaced 1K high-priority accounts responsible for 18.4bn in high-priority fraud exposure.

KEY PERFORMANCE INDICATORS

Top-level fraud metrics surfaced from 13 million transactions

18bn

Total Fraud Amount

18,42bn

High Priority Fraud Amount

0,06%

Recoverable Rate

1K

High Risk Accounts

18bn

Total fraud amount detected across all transactions

18.42bn

High-priority fraud flagged for immediate investigation

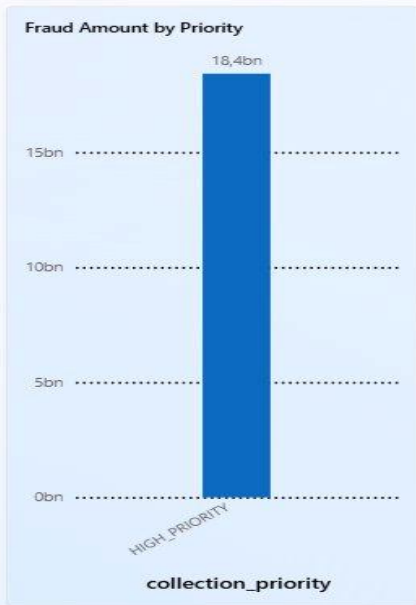
0.06%

Recoverable rate — most fraud is unrecoverable

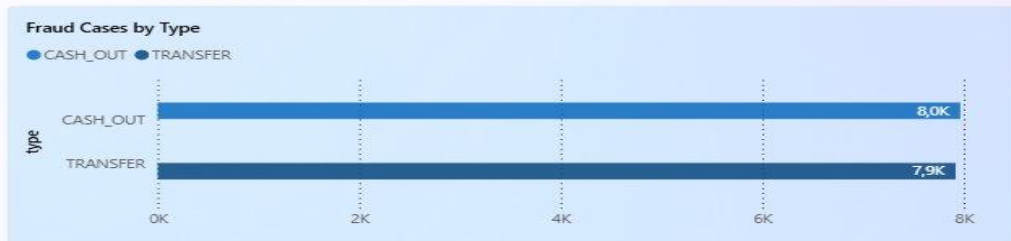
1K

High-risk accounts requiring urgent review

FRAUD OVERVIEW — AMOUNT, TYPE & RECOVERABILITY



💰 All 18.4bn in fraud is classified HIGH PRIORITY — no medium or low-risk buckets detected.



🏧 CASH_OUT (8K cases) and TRANSFER (7.9K) are the only fraud-linked transaction types.

📊 TRANSFER has a marginally higher recoverability (0.0015) vs CASH_OUT (0.0010) — both critically low.

HIGH-RISK ACCOUNT RANKING

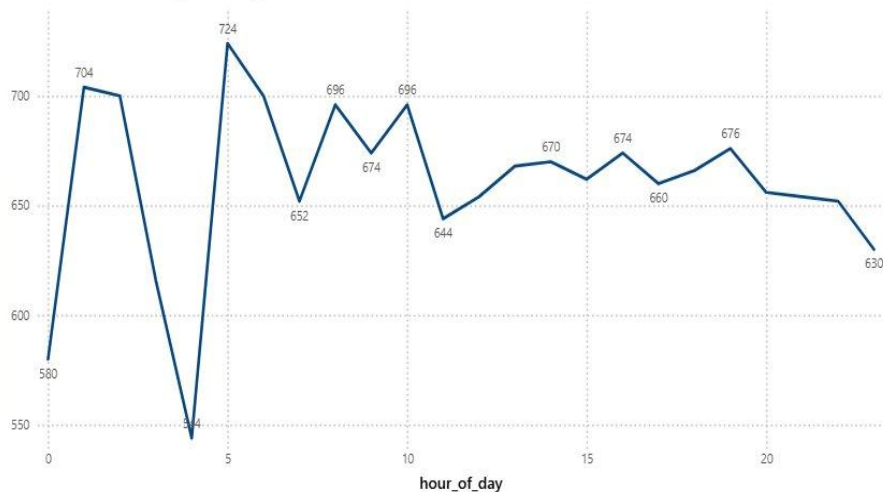
account_id	Sum of transaction_count	Sum of fraud_amount	Sum of fraud_count	Account Risk Rank
C429282035	2	19 993 773,28	2	1
C887830623	2	19 993 773,28	2	1
C154757729	2	19 955 522,10	2	2
C840795008	2	19 955 522,10	2	2
C1041938819	2	19 920 764,80	2	3
C427148786	2	19 920 764,80	2	3
C1071702423	2	19 775 638,12	2	4
C1548903046	2	19 775 638,12	2	4
C1983750370	2	19 622 208,98	2	5
C887693253	2	19 622 208,98	2	5
C1597398625	2	19 545 118,70	2	6
C417981063	2	19 545 118,70	2	6
C1040320190	2	19 498 085,90	2	7
C1139926828	2	19 498 085,90	2	7
C2078773097	2	19 451 674,16	2	8
C792281006	2	19 451 674,16	2	8
C1490771263	2	19 279 049,40	2	9
C560583390	2	19 279 049,40	2	9
C1452214555	2	19 278 105,66	2	10
C167488694	2	19 278 105,66	2	10
C1050497545	2	19 187 677,26	2	11
C1463520187	2	19 187 677,26	2	11
C452586515	2	19 170 080,74	2	12
C1372628002	2	18 936 128,10	2	13
C554608123	2	18 936 128,10	2	13
C851016578	2	18 931 977,64	2	14
C91276418	2	18 931 977,64	2	14
C1185250980	2	18 907 361,44	2	15
C1433301955	2	18 907 361,44	2	15
C476581036	2	18 843 713,74	2	16
C644898293	2	18 843 713,74	2	16
C1453976647	2	18 691 400,14	2	17
C200510021	2	18 691 400,14	2	17
Total	198	1 811 859 556,30	198	1

Key Findings

1. 198 accounts appear in the high-risk ranking, totalling 1.81 trillion in fraud exposure.
2. Top-ranked accounts each carry ~20bn in fraud across exactly 2 transactions — a strong indicator of large single-event fraud.
3. Risk Rank is calculated from combined fraud_amount and fraud_count, providing investigators a clear triage order.
4. Duplicate account IDs in the table suggest linked accounts sharing the same fraud event — a key network fraud signal.
5. All accounts have a transaction count of 2, implying fraudsters execute minimal activity to avoid pattern detection.

FRAUD TIMING ANALYSIS — HOUR OF DAY

Fraud Timing Analysis



Timing Insights

Hour 5

Peak fraud hour (724 cases) — early morning surge likely exploits low monitoring.

Hour 1

Second highest (704 cases) — overnight activity window.

Hour 4

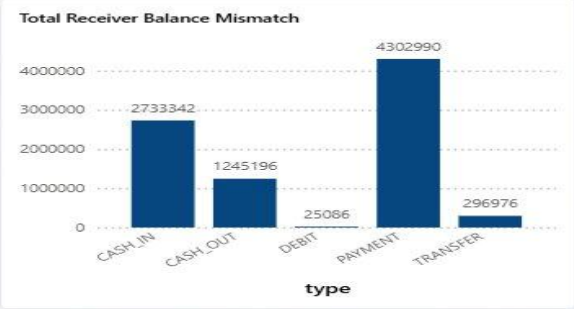
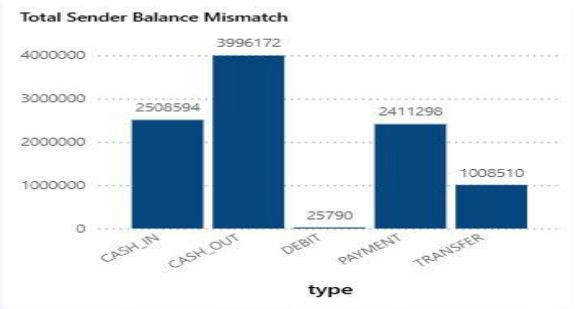
Dip to 544 cases just before the peak — possible preparation phase.

Hours 6–23

Activity stabilises at 630–700 range, suggesting persistent background fraud throughout the day.

💡 Recommendation: Increase automated monitoring intensity during hours 0–6 to intercept peak fraud windows.

BALANCE INTEGRITY — SENDER & RECEIVER MISMATCH



13M
Total Transactions

10M
Total Sender Mismatch

9M
Total Receiver Mismatch

67,61%
Receiver Mismatch Rate

78,19%
Sender Mismatch Rate

78.19%
Sender Mismatch Rate

67.61%
Receiver Mismatch Rate

10M
Total Sender Mismatches

9M
Total Receiver Mismatches

CASH_OUT
Highest Mismatch Type (both sides)

Summary & Recommendations

01 Focus on CASH_OUT & TRANSFER

These are the only two transaction types linked to fraud. Dedicated rule-sets and ML models for these types will maximise detection accuracy.

02 Increase Overnight Monitoring

Hours 1 and 5 show peak fraud activity. Automated alerting and staffed review during these windows should be prioritised.

03 Investigate Balance Mismatches

78% sender and 68% receiver mismatch rates signal systemic data integrity issues that may be masking additional fraudulent transactions.

04 Triage the 1K High-Risk Accounts

The ranked account table provides a ready-made investigation queue. Top accounts should be frozen pending manual review given 20bn+ exposure.